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À Votre Santé (B): Process Costing and Decision Analysis in the Wine Industry[1]

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BACKGROUND

Myka Freven was recently hired as manager of À Votre Santé, a small independent Napa-Green-certified winery located in the Napa Valley, California, American Viticultural Area (AVA) that is owned by her father, Jack Freven.[2] À Votre Santé (AVS), which means "to your health" in French, has a reputation for producing quality chardonnay wine. In 2005, Jack expanded the family's grape-growing operations and founded AVS as a means to involve his daughter in the wine business. Myka is a recent MBA graduate from the Haas School of Business at the University of California, Berkeley, and intends to take over the management of AVS in due time.

Being Napa-Green certified, the region's land use and environmental certification program, AVS is diligent about assessing cellar operations, implementing practices to conserve resources, improving comprehensive water and energy use, and conducting audits along with a waste diversion assessment.

AVS buys chardonnay grapes from the family's grape-growing operations and processes the grapes into a chardonnay-estate wine. Local restaurants purchase AVS wine to serve to their customers.

During her first few weeks as manager, Myka met with the managers and learned that the average demand and sales for the chardonnay estate was steady at 39,600 bottles of wine, and the average price was US$9.00 per bottle. The company, however, is experiencing declining net profits. Therefore, she decided to revisit the company's costing and pricing policy. She also shared with the managers her goal of maintaining a gross profit of 50% to cover AVS administrative expenses, which is consistent with operations of wineries. Myka met with the managers of the two production departments, the Crushing and Fermenting department and the Bottling department. From her meetings, she gathered that the company is using the weighted average method of preparing production reports under a process costing system. The managers in both departments consider all spoilage to be normal, and so, they ignore the costs of normal spoilage. This means no costs are added to goods that are spoiled, and spoilage is not reported because the cost of normal spoilage is included in the cost of good units completed.

Furthermore, the managers noted that they have worked hard to maintain the appropriate control measures and reduce their abnormal spoilage to 0% in the last five years. So, they inspect goods only at the end of the process in each department.

WINEMAKING PROCESS

Harvest takes place in the late summer and early fall months; typically, the time elapsed from harvest to final sale is about 11 months. The chardonnay grapes are purchased from AVS Vineyards at US$1.34 per pound, instead of purchasing from an outside vineyard. An important quality of Chardonnay wine is dependent upon the acidity of the grapes. So the grapes are picked from AVS Vineyards two weeks earlier than other producers. AVS Vineyard sells to other wineries in the market at US$1.34 per pound, and transfers to the AVS winery at the same market price of US$1.34 per pound.[3] On average a purchase of 40 pounds of grapes are required to produce one case (12 bottles) of wine. In August 2017, AVS purchased 382,836 pounds of grapes from AVS vineyards at US$1.34 per pound. The following are the major processes and departments in the winery:

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CRUSHING AND FERMENTING

Crushing: The grapes are transferred to the winery for washing and crushing after harvest. The crushing process separates the juice from the pulp, skin, and stems. The juice is used to make the wine; the pulp, skin, and stems are recycled as compost onto the fields whenever possible or disposed of according to the Napa Green certified land program.

Fermenting: The extracted juice is initiated into the fermenting process. An important factor in the final taste of chardonnay is whether the juice is aged in oak. Chardonnay is fermented using oak barrels, which gives flavor to the final product. White wines have a shorter timeline for aging, and they are barrel-aged because they are meant to be consumed within five years. All fermenting takes place in a temperature-controlled environment; however, each fermenting method results in some wine loss through evaporation. The expected loss in volume through fermentation is 10% and is considered to be normal spoilage.[4] At the end of this process, each barrel of wine is one *unit*. The department manager classifies all spoilage as normal, related to loss from fermentation.

Bottling: Approximately 200 pounds of crushed grapes are filled in a barrel of wine. Each barrel transferred to the bottling department can yield 11 cases or 132 bottles of wine. Bottling is a very important process, and AVS takes every precautionary measure, such as proper sanitation, cleaning, bottle washing, controlling sulfur dioxide depletion, and inspecting in order to prevent contamination and inefficient operations. Spoilage is 10%, and the accountant and managers have determined that all spoilage is considered normal and is detected at the end of the production process. They insist that spoilage in the Bottling department is related to the technology of the machines used in the facility. Since AVS does not have state-of-the-art technology, there is some spoilage related to bottle cracking or breaking during the filling process.[5] The managers believe that this spoilage is unavoidable. They are considering making an investment in two years to update the technology in the Bottling department. Tables 1 and 2 show the information for August 2017.

Table 1: Data for the Crushing and Fermenting Department for August 2017

	Physical units (barrels)	Direct materials	Conversion costs
Work in process, August 1	200	$68,400	$58,120
Degree of completion of beginning work in process		100%	70%
Started during August 2017	1,500		
Completed during August 2017	1,200		
Work in process, August 31	500		
Degree of completion of ending work in process		100%	60%
Total costs added during August 2017		$513,000	$436,000

Note: All monetary values are in US dollars.

Table 2: Data for the Bottling Department for August 2017

	Physical units (barrels)	Transferred-in	Direct materials	Conversion costs
Work in process, August 1	250	$100,000	$0	$7,105
Degree of completion of beginning work in process		100%	0%	70%
Transferred-in during August 2017	?			
Completed during August 2017	300			
Work in process, August 31	1,150			
Degree of completion of ending work in process		100%	0%	60%
Total costs added during August 2017		?	$18,400	$14,025

Note: All monetary values are in US dollars.

REQUIREMENTS:

1. **Production Report Analysis:** Prepare production reports under the weighted average as well as the FIFO method. What is the cost per bottle under each method? What is considered as normal and abnormal spoilage, respectively in the wine making process? How are the costs associated with normal and abnormal spoilage treated? Is it appropriate to ignore the cost of normal spoilage because all spoilage is normal?

2. **Decision Analysis:** Myka would like to maintain a 50% gross profit to cover AVS's administrative expenses of US$125,000 and target a 15% net profit. What should the target cost per case be in order to achieve these goals? Does the current production report in AVS help achieve these targets? Does your final production report show costs per case that will lead to achieving these targets?

3. **Ethical Implications:** What are the possible reasons for managers to state that all spoilage is normal? What additional information would you request from managers to confirm that there is no abnormal spoilage and that all spoilage is normal?

ENDNOTES

[1] "À Votre Santé (B): Process Costing And Decision Analysis In The Wine Industry" is an extension of Priscilla S. Wisner, "À Votre Santé: Product Costing and Decision Analysis in the Wine Industry," *IMA Educational Case Journal*, June 2011. www.imanet.org/-/media/08756f85afc34ffd9c23b33ba5dfba4b.ashx.

[2] AVAs are wine grape-growing areas established by the Department of the Treasury's Alcohol and Tobacco Tax and Trade Bureau. AVAs are established to help vintners better describe the origin of their wines. As of December 2018, there are 242 AVAs in the United States; the Napa Valley AVA consists of 16 sub-AVAs, each determined by geographical boundaries.

[3] "The average market price is from the 2016 grape crush report published in March 2017. Chardonnay average price per ton is US$2,673 in 2016. Commonly, grape purchase agreements between growers and wineries use the weighted average price per ton as a baseline for setting the contracted price per ton. The high and low per ton of chardonnay grape prices in 2016 were US$12,574 and US$400, respectively. Wine grape prices and production were up throughout California in 2016. The supply levels of grapes and the amount of juice generated from the grapes are dependent on the yearly climatic conditions and other factors, such as temperature, rootstock, and fertilizers used. For more, see: 2017 Report – Sonoma and Napa Premium Wine Grape Prices 2016, Vintroux, March 2017, http://vineyardandwinerysales.com/blog/2017-report-sonoma-napa-premium-wine-grape-prices-2016/.

[4] Bacterial spoilage could also occur in the fermentation stage, because oak barrels are a specific source of bacteria. The correct operational protocol with sterile filtration procedures followed by the addition of sufficient sulfur dioxide should prevent potential spoilage from bacteria. The department managers have assessed that the procedures in the Crushing and Fermenting department follow the correct protocols to prevent such inefficiencies resulting in spoilage.

[5] The industry is moving toward low-weight glass bottles with resistance to breakage. AVS purchases the best quality glass bottles but does not have state-of-the-art technology to fill the bottles. The pressure the glass encounters from the filling machine during the filling process leads to breaking of bottles, which is unavoidable.

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